Mufeng Chen

Ph.D. Student and Graduate Research Assistant, Electrical and Computer Engineering
chen5240@purdue.edu | West Lafayette, IN
GitHub: github.com/chenmufeng0119

SUMMARY

- Research on memory systems and architecture for AI, including compute-in-memory and hybrid memory systems.

EDUCATION

Purdue University — Ph.D. Student
Electrical and Computer Engineering | West Lafayette, IN, USA | 2024–present
- Advised by Prof. Haitong Li in the NanoX Lab.
- Research interests include AI memory systems, compute-in-memory, and hybrid memory systems.

Zhejiang University — Graduate Study
Electronic Engineering | Hangzhou, China | 2021–2024

Huazhong University of Science & Technology — B.Eng.
Optical and Electronic Information Engineering | Wuhan, China | 2017–2021

EXPERIENCE

Graduate Research Assistant, NanoX Lab — Purdue University
West Lafayette, IN, USA | 2024–present
Advised by Prof. Haitong Li.

Research Assistant, RFNE Research Center — Zhejiang University
Hangzhou, China | 2021–2024
Advised by Prof. Er-Ping Li.

RESEARCH INTERNSHIPS

Research Intern, NanoX Lab — Purdue University
West Lafayette, IN, USA | 2023–2023
Summer internship, advised by Prof. Haitong Li.
- Research interests included compute-in-memory and hyperdimensional computing.

Research Intern, SIMS Lab — Rice University
Houston, TX, USA | 2022–2022
Summer internship, advised by Prof. Kaiyuan Yang.

EARLIER EXPERIENCE

Research Assistant — Wuhan National Laboratory for Optoelectronics
Wuhan, China | 2019–2019
Advised by Prof. Chen Lin.

SELECTED PUBLICATIONS

- M. Chen et al. “Analog Multilevel eDRAM-RRAM CIM for Zeroth-Order Fine-tuning of LLMs.” 2025 IEEE International Memory Workshop, pp. 1–4
- L. Zheng et al. “PROTEUS: A 40 nm Programmable General-Purpose Digital Compute-In-Memory Accelerator With eNVM and Hierarchical ISA for Versatile Edge AI.” IEEE Journal of Solid-State Circuits, 2026
- L. Zheng et al. “Heterogeneous Compute-in-Memory Fabrics for Efficient, Scalable Edge Inference and Learning.” 2026 International Symposium on Quality Electronic Design
- L. Zheng et al. “CENTAUR: A 38.5-TFLOPS/W 600MHz Floating-Point Digital Compute-In-Memory Engine with 40nm Fusion RRAM-eDRAM Macros Featuring 3D-MAC Operation.” 2025 IEEE Asian Solid-State Circuits Conference
- E.-P. Li et al. “An Electromagnetic Perspective of Artificial Intelligence Neuromorphic Chips.” Electromagnetic Science, 1(3), 2023
- C. Yu et al. “MAP-SNN: Mapping Spike Activities with Multiplicity, Adaptability, and Plasticity into Bio-Plausible Spiking Neural Networks.” Frontiers in Neuroscience, 16, 945037, 2022

TECHNICAL SKILLS

Circuit & EDA: Cadence Virtuoso, Synopsys VCS, Design Compiler, Innovus

Programming: Python (PyTorch), C/C++, MATLAB, Verilog/SystemVerilog

Research areas: AI memory systems and architecture, compute-in-memory, hybrid memory systems, High-Bandwidth Flash research, hyperdimensional computing